

Protective effect of myo-inositol hexaphosphate (phytate) on bone mass loss in postmenopausal women.

INTRODUCTION: The objective of this paper was to evaluate the relationship between urinary concentrations of InsP6, bone mass loss and risk fracture in postmenopausal women. MATERIALS AND METHODS: A total of 157 postmenopausal women were included in the study: 70 had low ($\leq 0.76 \mu\text{M}$), 42 intermediate ($0.76\text{--}1.42 \mu\text{M}$) and 45 high ($\geq 1.42 \mu\text{M}$) urinary phytate concentrations. Densitometry values for neck were measured at enrollment and after 12 months (lumbar spine and femoral neck), and 10-year risk fracture was calculated using the tool FRAX®. RESULTS: Individuals with low InsP6 levels had significantly greater bone mass loss in the lumbar spine ($3.08 \pm 0.65 \%$ vs. $0.43 \pm 0.55 \%$) than did those with high phytate levels. Moreover, a significantly greater percentage of women with low than with high InsP6 levels showed more than 2 % of bone mass loss in the lumbar spine (55.6 vs. 20.7 %). The 10-year fracture probability was also significantly higher in the low-phytate group compared to the high-phytate group, both in hip ($0.37 \pm 0.06 \%$ vs $0.18 \pm 0.04 \%$) and major osteoporotic fracture ($2.45 \pm 0.24 \%$ vs $1.83 \pm 0.11 \%$). DISCUSSION: It can be concluded that high urinary phytate concentrations are correlated with reduced bone mass loss in lumbar spine over 12 months and with reduced 10-year probability of hip and major osteoporotic fracture, indicating that increased phytate consumption can prevent development of osteoporosis.